

MongoDB CRUD Quick Notes (Library Example)

1. Database Commands

```
show dbs
use libraryDB
db
db.dropDatabase()
```

2. Collection Commands

```
show collections
db.createCollection("books")
db.books.drop()
```

3. Insert (Create)

```
db.books.insertOne({ title: "DBMS", author: "Korth", available: true })

db.books.insertMany([
  { title: "DSA", author: "Mark" },
  { title: "Java", author: "James" }
])
```

4. Read (Find)

```
db.books.find()
db.books.find().pretty()
db.books.find({ author: "Korth" })

db.books.find({ price: { $gt: 500 } })
db.books.find({ price: { $lt: 500 } })
db.books.find({ author: { $ne: "Korth" } })

db.books.find({ author: "Korth", available: true })
db.books.find({ $or: [{ author: "Korth" }, { available: true }] })
```

5. Update

```
db.books.updateOne(
  { title: "DBMS" },
  { $set: { available: false } }
)

db.books.updateMany(
  { available: true },
  { $set: { available: false } }
)

db.books.updateOne(
  { title: "DBMS" },
  { $inc: { price: 50 } }
)
```

6. Delete

```
db.books.deleteOne({ title: "Java" })  
db.books.deleteMany({ available: false })
```

7. Sorting, Limit, Count

```
db.books.find().sort({ price: 1 })  
db.books.find().sort({ price: -1 })  
db.books.find().limit(2)  
db.books.countDocuments()
```

8. Advanced Queries

```
db.books.find({ issuedTo: null })  
db.books.find({ issuedTo: { $ne: null } })  
db.books.find({ price: { $exists: true } })  
db.books.find({ title: /data/i })
```

9. Projection

```
db.books.find({}, { title: 1, author: 1 })
```

10. Key Differences

Table → Collection

Row → Document

Column → Field

Viva Tip: MongoDB CRUD operations include insertOne, find, updateOne, and deleteOne. It also supports filtering, sorting, and operators like \$gt, \$or, and \$ne.